

## Explain this project end to end.

DermIntel is a personalized skincare analysis platform built to help users decide whether a product is a good fit for their own skin profile rather than relying on generic beauty advice. The system starts when a user signs in, completes a one-time onboarding flow, and creates a persistent skin passport containing details such as skin type, sensitivity, concerns, goals, allergies, and ingredients they prefer to avoid.

Once the profile is saved, the user can paste a product URL or submit an ingredient list manually. If a URL is provided, the backend begins a product-verification pipeline. It validates that the link is a real product page, fetches the page content, extracts product metadata such as brand and product name, and searches for ingredient candidates from retailer content, structured metadata, embedded product JSON, and other machine-readable sections on the page.

If the ingredient list is not clearly available from the retailer page, the system expands the search to additional trusted sources such as official brand pages, recognized cosmetic databases, distributor pages, and product-label image fallback. Every candidate list is checked to ensure it looks like a real full INCI formula rather than marketing copy, hero ingredients, or benefit claims. If no trustworthy formula can be verified, the system stops instead of guessing.

After the formula is verified, the ingredient-analysis pipeline begins. Each ingredient is normalized and mapped to DermIntel's ingredient knowledge base, which stores purpose, benefits, suitable skin types, avoid skin types, irritation signals, comedogenicity, concern mappings, and supporting references. The system also estimates each ingredient's likely influence from its position in the INCI order, because earlier ingredients usually have greater impact on the formula.

The scoring engine then compares the verified formula against the user's skin passport. Safety is calculated by looking at factors such as irritation risk, allergen overlap, comedogenicity, and sensitivity-related penalties. Suitability is calculated by checking how strongly the ingredients support or conflict with the user's goals and concerns, such as hydration for dry skin, oil control for oily skin, or soothing support for sensitive skin.

The final result is returned to the dashboard as a personalized verdict. The user sees overall safety and suitability scores, positives, watchouts, an ingredient-by-ingredient breakdown, estimated concentration ranges based on INCI order, and a natural-language explanation of why the product may or may not suit their profile. In short, DermIntel combines product verification, ingredient intelligence, and profile-based scoring into one end-to-end decision-support experience for skincare selection.